

Scenario 1 - Buffer Full, Producer Waiting:

```
((base) matthewbosworth@Matthews-Laptop-4 Desktop % cd OS-Group-Assignment1-Java
((base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-Java % javac ProducerConsumer.java
(base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-Java % java ProducerConsumer
=====
  PRODUCER-CONSUMER PROBLEM DEMONSTRATION
  Buffer Size: 5
=====

*** SCENARIO 1: Buffer Full - Producer Waiting ***
Producer will produce 7 items, Consumer will consume 7 items
Producer is faster, so buffer will fill up

Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 1. Count = 2
Buffer: [FULL, FULL, EMPTY, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 1
Buffer: [EMPTY, FULL, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 0. Count = 2
Buffer: [FULL, FULL, EMPTY, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 1
Buffer: [EMPTY, FULL, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 0. Count = 2
Buffer: [FULL, FULL, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 2. Count = 3
Buffer: [FULL, FULL, FULL, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 2
Buffer: [EMPTY, FULL, FULL, EMPTY, EMPTY]
-----
Consumer: Consumed at index 1. Count = 1
Buffer: [EMPTY, EMPTY, FULL, EMPTY, EMPTY]
-----
Consumer: Consumed at index 2. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----

*** Scenario 1 Complete ***
```

Scenario 2 - Buffer Empty, Consumer Waiting:

```
*** SCENARIO 2: Buffer Empty - Consumer Waiting ***
Consumer starts first and tries to consume from empty buffer

Consumer: Buffer is EMPTY. Waiting...
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----

*** Scenario 2 Complete ***
```

Scenario 3 - Buffer Partially Full

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*** SCENARIO 3: Partially Full Buffer ***
Producer and Consumer working at similar speeds
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Producer: Produced at index 0. Count = 1
Buffer: [FULL, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
Consumer: Consumed at index 0. Count = 0
Buffer: [EMPTY, EMPTY, EMPTY, EMPTY, EMPTY]
-----
```

```
*** Scenario 3 Complete ***
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ALL SCENARIOS COMPLETED SUCCESSFULLY!
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